Session 3: Let the loop do the walking

Question: Can one rule label every order line?

Concepts

Concept	In daily life	In Excel	The limit
for loop	Walking down the shelf, checking each bin	Dragging a formula down	The body must be indented
function	Your own reusable rule	Your own named formula	It does nothing until you call it
return	The answer the rule hands back	The value the formula shows	print only shows. return hands it back
traceback	A stack of plates, newest on top	None	Read the bottom line first, then go up

Code patterns

```
for quantity in quantities:
    print(quantity)

def order_size(quantity):
    if quantity >= 100:
        return "big"
    elif quantity >= 12:
        return "medium"
    else:
        return "small"

for quantity in quantities:
    print(f"{quantity} -> {order_size(quantity)}")
```

Three common errors

Error	What to do
IndentationError	The lines inside the loop or function are not indented.
None instead of an answer	The function uses print instead of return.
NameError for a function	You called the function before you ran the cell with `def`.

Exercises and expected answers

Exercise	Expected
Predict: How many times does done print? Guess first.	Guess first, then run it to check
Predict: This function prints but does not return. What is saved in answer? Guess first.	Guess first, then run it to check
Fill in: Label every quantity in the list.	6 units -> small and so on, 7 lines
Fill in: A function that returns the value of a line, rounded to 2 decimals.	15.3
Fix it: The loop body is not indented. Fix it so it prints the 7 quantities.	The cell runs without an error
Fix it: This is a real traceback: one quantity is text. Read it from the bottom, then fix the list.	small, medium, small
Write it: Loop over the five lines and print the value of each with line_value.	15.3, 20.34, 22.0, 20.34, 20.34
Write it: Change order_size so medium starts at 50, then label 6, 12, 99, 100 and 1930.	small, small, medium, big, big
Write it: Write is_uk(country) that returns True for United Kingdom and False for any other country.	True, False
Case: Invoice 540815 is a real big order with three lines: 1930 units at 2.55, 2000 at 1.85 and 3114 at 2.10. For each line, print its size and its value.	1930 units: big, 4921.5 pounds, then 3700.0 and 6539.4
Extra: Write your own function with a return: with_tax(price) gives the price plus 20%, rounded to 2 decimals.	3.06
Extra: Before Session 4: open a blank Colab notebook, run these two lines once and accept the Google permission screen.	Your own